

Outlier removal using the IQR method

As a civil engineer, you are analyzing the dataset, and believe that there are outliers in compressive strength that need to be removed before the dataset can be used for model training.

Please identify these outliers using the **IQR method**.

* The compressive strength values can be imported using the code below.

```
T = readtable("Concrete_Data.xls", 'VariableNamingRule','preserve');
values = table2array(T(:,9));
```

Script

Save

Reset

MATLAB Documentation (https://www.mathworks.com/help/)

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```
2 values = table2array(T(:,9));
3 % Visualize the data here if necessary
4 histogram(values)
5 xlabel('compressive strength');
6 ylabel('Number of samples');
7
8 X = values;
9 % Filtering out those samples lies outside the IQR
10 % identify IQR, your code here:
11 Q1 = prctile(X, 25);
12 Q3 = prctile(X, 75);
13 IQR = Q3 - Q1;
14 % Remove outliers, your code here:
15 X = X(X < Q3 + 1.5*IQR & X > Q1 - 1.5*IQR);
16 % Visualize again
17 histogram(X)
18
```

Run Script

Tests: All Tests Passed (100%)

Submit

Keep samples that are not outliers	34% (34%)
Outlier test #1	33% (33%)
Outlier test #2	33% (33%)

Total: 100%

Output

